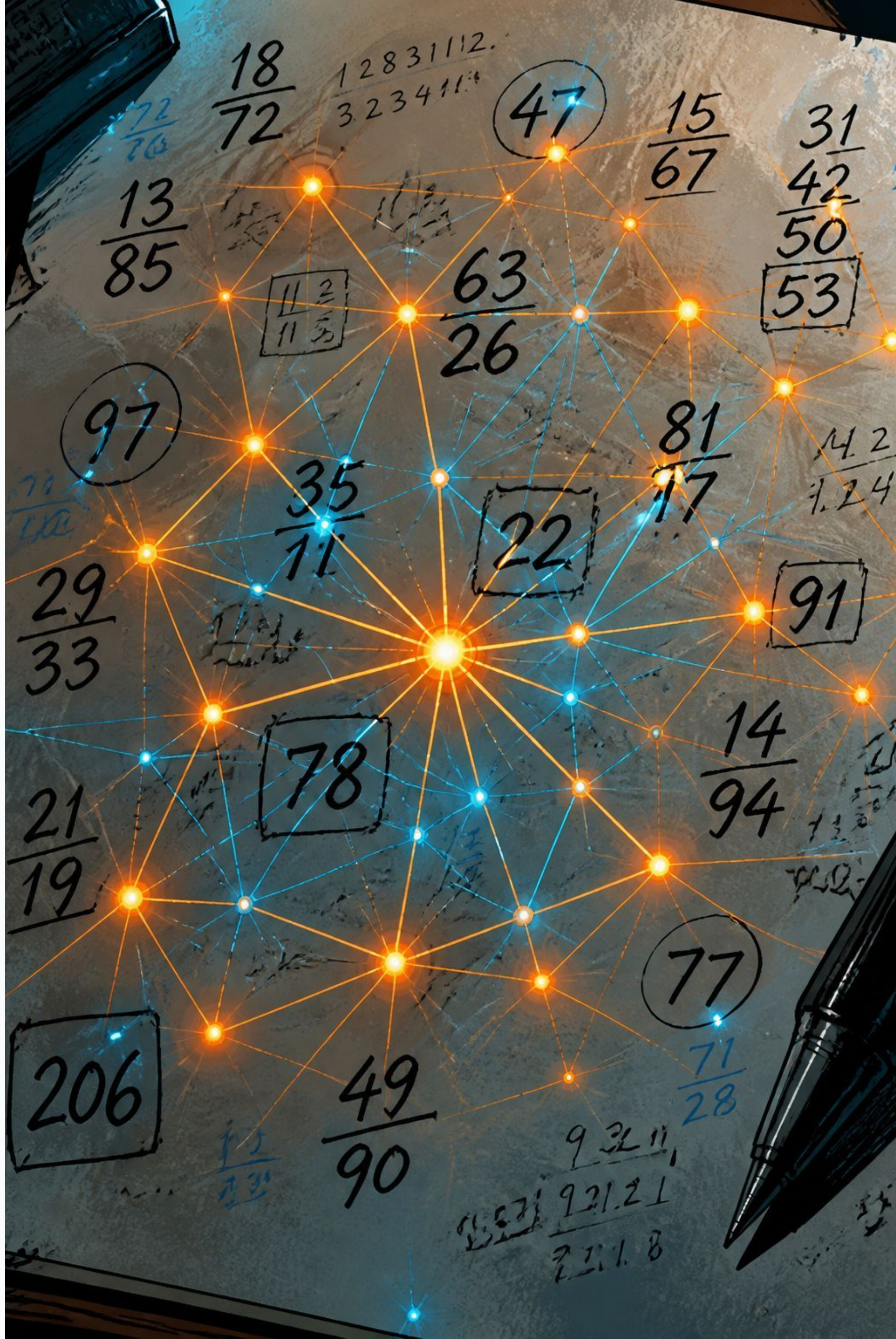
# ✍️ Handwritten Digit Recognition

A web application that uses deep learning to recognize handwritten digits (0–9), built with TensorFlow, Keras, and Streamlit.

AI / ML | COMPUTER VISION | OPEN SOURCE



# What It Does

### Upload Images

Upload any handwritten digit image through a simple web interface.

### Instant Prediction

The model predicts which digit (0–9) is in the image in real time.

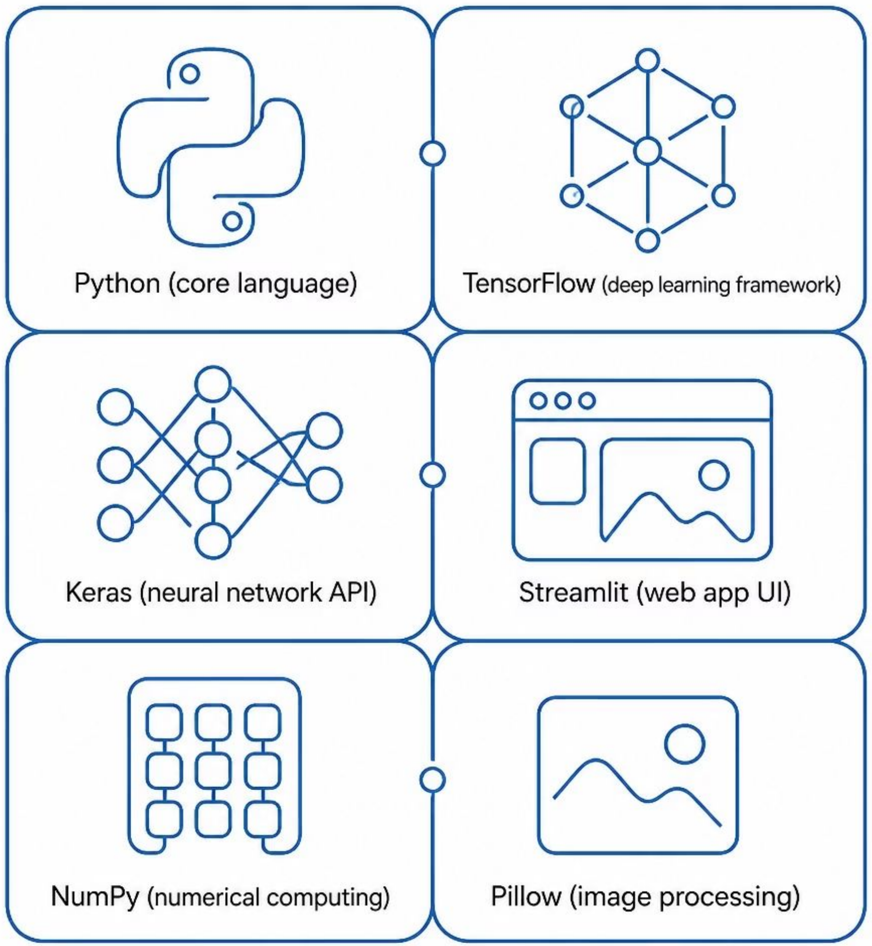
### Confidence Score

Displays a confidence score showing how certain the model is about its prediction.

# Trained on MNIST

The model is trained using the **MNIST dataset** — the gold standard benchmark for handwritten digit recognition. MNIST contains 70,000 labeled images of handwritten digits, making it ideal for training and evaluating digit classification models.

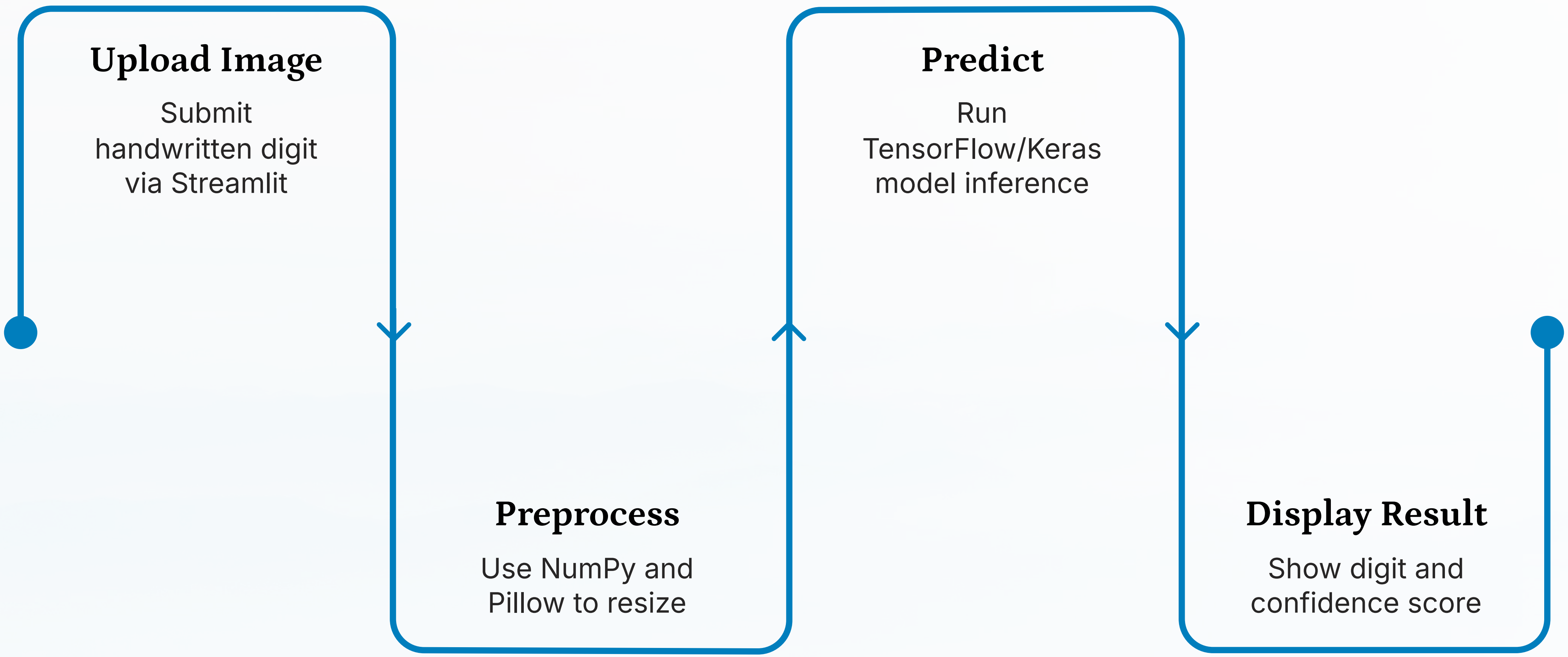
# Tech Stack




Six tools work together to power the application:

- **Python** — core programming language
- **TensorFlow** — deep learning framework for building and training the neural network
- **Keras** — high-level neural network API running on top of TensorFlow
- **Streamlit** — turns Python scripts into interactive web apps
- **NumPy** — numerical computing and array operations
- **Pillow** — image loading and processing

# How It Works




The pipeline flows from image upload through preprocessing, model prediction, and result display. Users upload a handwritten digit image through the Streamlit interface. NumPy and Pillow preprocess the image for the model. The TensorFlow/Keras neural network then classifies the digit (0–9) and returns a confidence score, all displayed instantly in the browser.


# Run Locally

Getting the app running on your machine takes just three commands. First, install all required packages listed in requirements.txt. Then run train_model.py to train the TensorFlow/Keras model on the MNIST dataset. Finally, launch the Streamlit web interface with app.py.

```
pip install -r requirements.txt
python train_model.py
python -m streamlit run app.py
```

# Project Structure

### `app.py`

The Streamlit web application — handles image upload, runs predictions, and displays results with confidence scores.

### `train_model.py`

Script that trains the TensorFlow/Keras model on the MNIST dataset and saves it for use by the app.

### `requirements.txt`

Lists all Python dependencies: TensorFlow, Keras, Streamlit, NumPy, and Pillow.

# Deep Learning Under the Hood

TensorFlow and Keras power the core recognition engine. The model learns to identify patterns in handwritten digits through training on thousands of labeled MNIST examples, then applies those learned patterns to classify new, unseen images with a confidence score.

# Key Features at a Glance

### Image Upload

Simple drag-and-drop or file-select interface for uploading handwritten digit images.

### AI Prediction

TensorFlow/Keras model trained on MNIST classifies digits 0–9 instantly.

### Confidence Score

Shows how confident the model is in each prediction, giving users transparency into results.

### Web Interface

Streamlit provides a clean, browser-based UI — no complex setup needed for end users.

# Get Started

This is an open-source project by **ArchitVaksh** on GitHub. Clone the repo, install dependencies, train the model, and launch the app in minutes.

### 📂 Clone the Repo

github.com/ArchitVaksh/handwritten-digit-recognition

### ⚙️ Install & Run

Follow the three-step local setup to get the app running.

### 🚀 Explore & Extend

Fork, modify, and build on top of this MNIST-based digit recognizer.